

Secure Military Messaging App with Local-Only Storage and End-to-End Encryption

Mini Project Report

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Abstract

Building a simple secure messaging prototype that demonstrates how private communication can be safeguarded without depending on server-side storage is the main goal of this small project. Nowadays, the majority of messaging apps store user messages or logs on centralized servers, which creates a risk in the event that those servers are compromised or monitored. The objective of this project is to develop an intuitive end-to-end encrypted chat system in which all message data is stored entirely on the user's device and is promptly deleted from the server upon delivery. This method keeps previous conversations from being retrieved by outsiders and lessens the possibility of data leaks. The mini project primarily demonstrates the fundamental idea of encrypted communication and lays the groundwork for more sophisticated features that will be developed in subsequent phases, including metadata protection and self-destructive messages.